

Bespoke Agentic AI Factory

Composable · Observable · Governed AI pipelines for the enterprise

Agentic RAG **AWS Kiro** **AWS Transform** **Spring AI** **DeepEval Gates**

Solution Architecture · Pipeline Flows · Live Demo

35-slide demo pack

One factory, five governed stages

- Turn scattered enterprise knowledge into grounded, cited answers
 - Every answer passes automated quality + safety gates before release
 - Modernise legacy Java/Spring estates with agentic transforms
 - Full observability: metrics, traces and a single pane of glass
 - Ships on AWS (ECS Fargate) via Terraform + Harness with canary releases
- 5

governed pipeline stages
- 2

internal sources wired (Confluence, SharePoint)
- 99

automated tests, all green
- 4

deploy environments w/ canary

Enterprise AI stalls on trust, silos and legacy

Knowledge silos

- Docs spread across Confluence,
- SharePoint, wikis and drives
- No unified, searchable context

Unverified output

- LLMs hallucinate and drift
- No automated quality gate
- Compliance / audit risk

Legacy estates

- Old Spring Boot, CVE exposure
- Hardcoded rule engines
- Manual, slow modernisation

→ **The gap: a governed, observable pipeline that grounds, verifies and modernises — not just a chatbot.**

Capabilities at a glance



Agentic RAG

Hybrid vector + graph,
multi-hop retrieval



Source connectors

Confluence & SharePoint
→ one RAG index



Eval gates

DeepEval metrics +
LLM-as-judge, CI-gated



AWS Transform

7 modernisation
definitions (atx)



Spring AI

Boot 2.x → 3 +
Spring AI 1.0



Single pane of glass

Metrics, traces,
RED per stage



Secure delivery

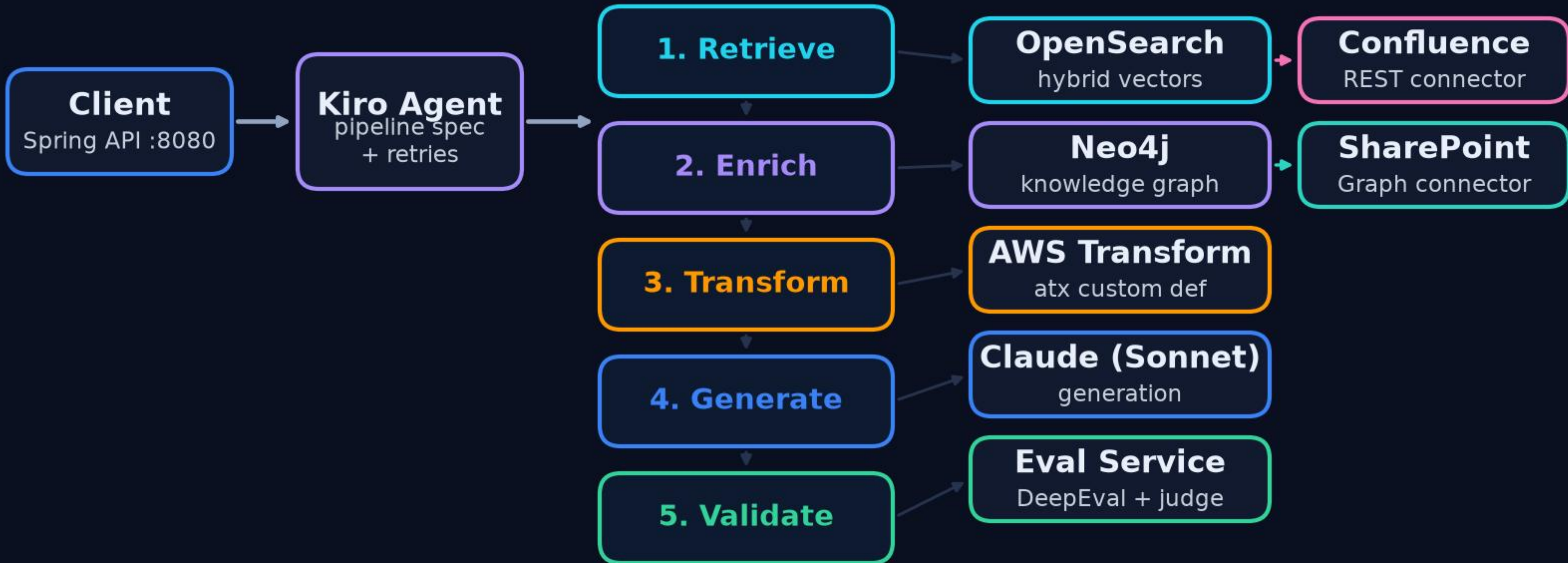
SAST/DAST gates,
canary releases



Knowledge graph

Neo4j entity
enrichment

High-level system architecture



OBSERVABILITY

OTel Collector **Prometheus** **Jaeger** **Grafana** **CloudWatch**

AWS · ECS Fargate · ALB · Terraform · Harness

CORE FLOW

The 5-stage agentic pipeline

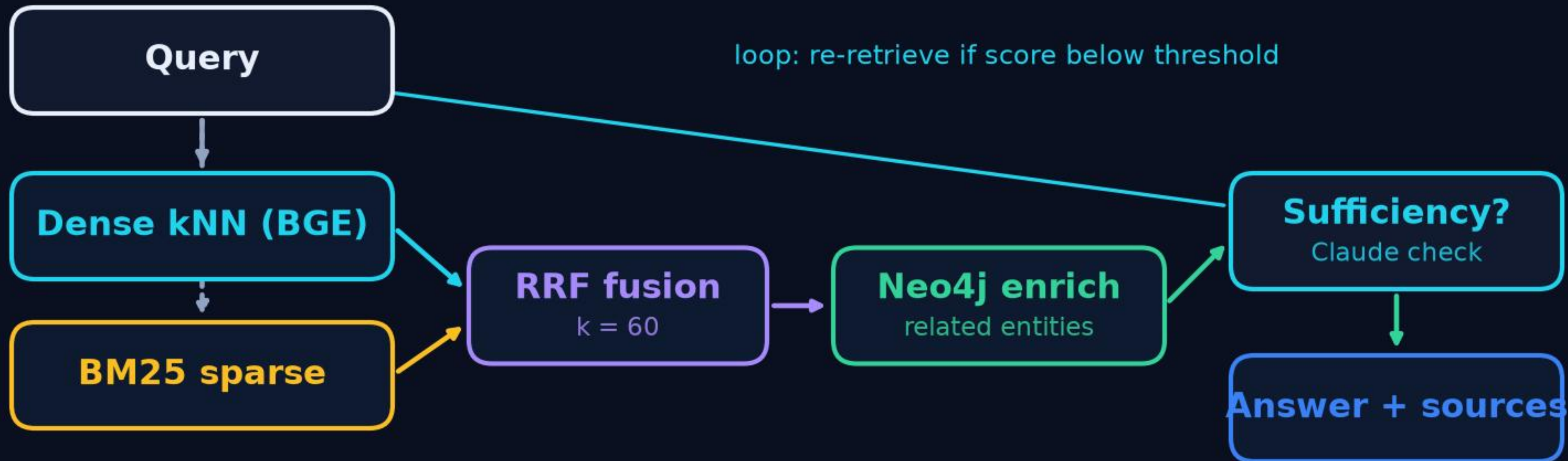
multi-hop: re-retrieve until sufficiency met



Why it matters

- Each stage is an independent, instrumented service (OTel span per stage)
- Retry + sufficiency scoring make retrieval agentic, not single-shot
- The validate stage can block a bad answer before it reaches the user

Agentic RAG — hybrid retrieval + graph



Hybrid

dense + sparse, best of both

Grounded

every claim cites a source

Agentic

multi-hop with sufficiency

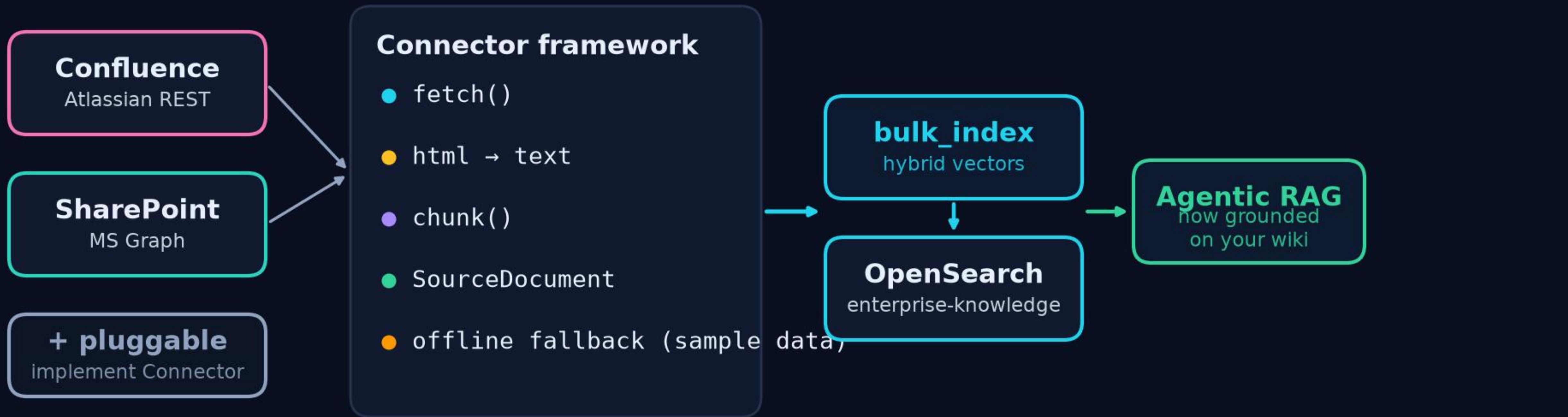
Filtered

metadata + dept scoping

NEW CAPABILITY

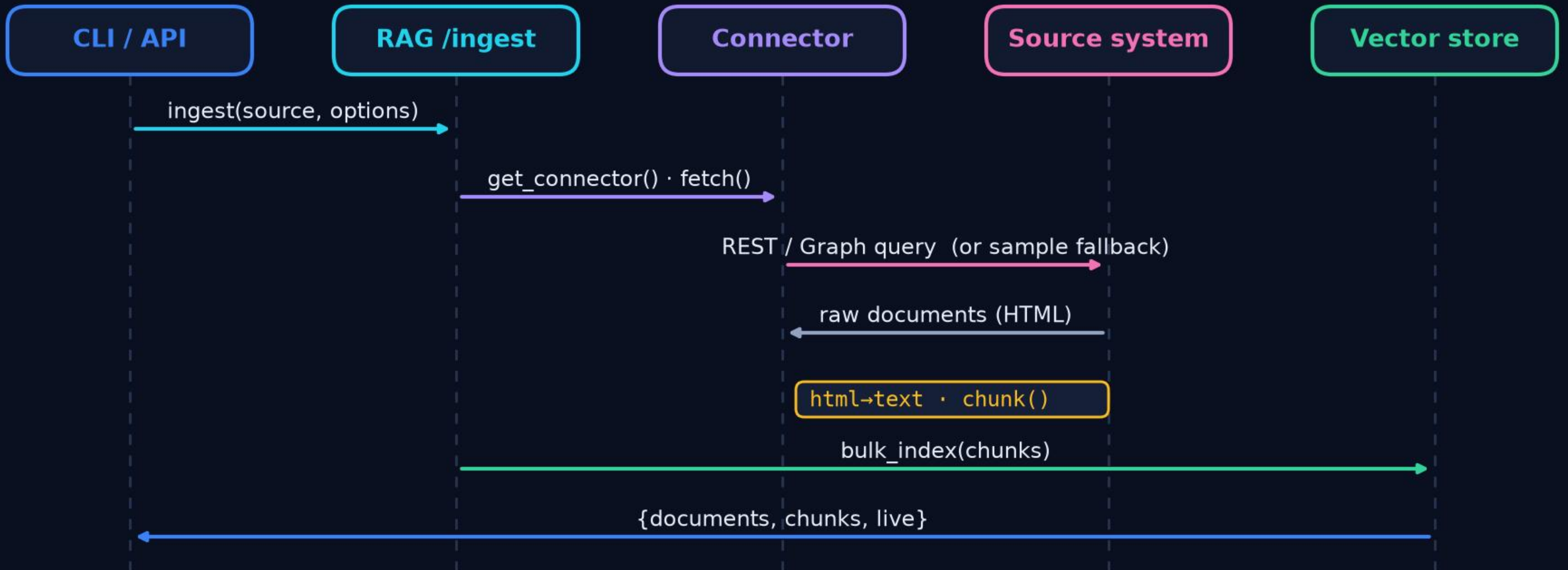
Internal source connectors

Confluence & SharePoint → the agentic RAG index



`POST /ingest {"source":"confluence"}` · `GET /sources` · `python scripts/ingest_sources.py --source sharepoint`

Connector ingestion — sequence



Offline-safe: with no credentials the connector returns bundled sample pages, so the demo always runs.

Evaluation gates — quality before release

Answer relevanc ≥ 0.80
threshold ≥ 0.80

Faithfulness ≥ 0.85
threshold ≥ 0.85

Contextual recal' ≥ 0.70
threshold ≥ 0.70

Hallucination ≤ 0.10
threshold ≤ 0.10

Toxicity ≤ 0.05
threshold ≤ 0.05

Bias ≤ 0.10
threshold ≤ 0.10

LLM-as-judge

- Weighted rubric across metrics
- Self-consistency sampling
- Returns weighted_score + pass/fail
- Runs per answer (validate stage)
- Same suite gates CI deployment

PASS → release · **FAIL** → block + log

AWS Transform Custom — the factory

enterprise-context-normaliser

PII redaction · date norm · truncation

java-spring-to-spring-ai

legacy Spring → Spring AI 1.x

eval-dataset-updater

regenerate eval cases on schema change

fintech-java-spring2-to-spring3

Boot 2.x → 3.2 · WebClient · OTel

fintech-rules-to-agentic-rag

rule engines → Kiro agentic RAG

fintech-soap-to-rest

SOAP → REST anti-corruption layer

fintech-cve-remediation

scan → map CVE → patch, gated to 0 HIGH

Driven by AWS Kiro agent specs · 6-service fintech legacy estate as the modernisation demo

Spring AI modernisation

LEGACY · Spring Boot 2.7

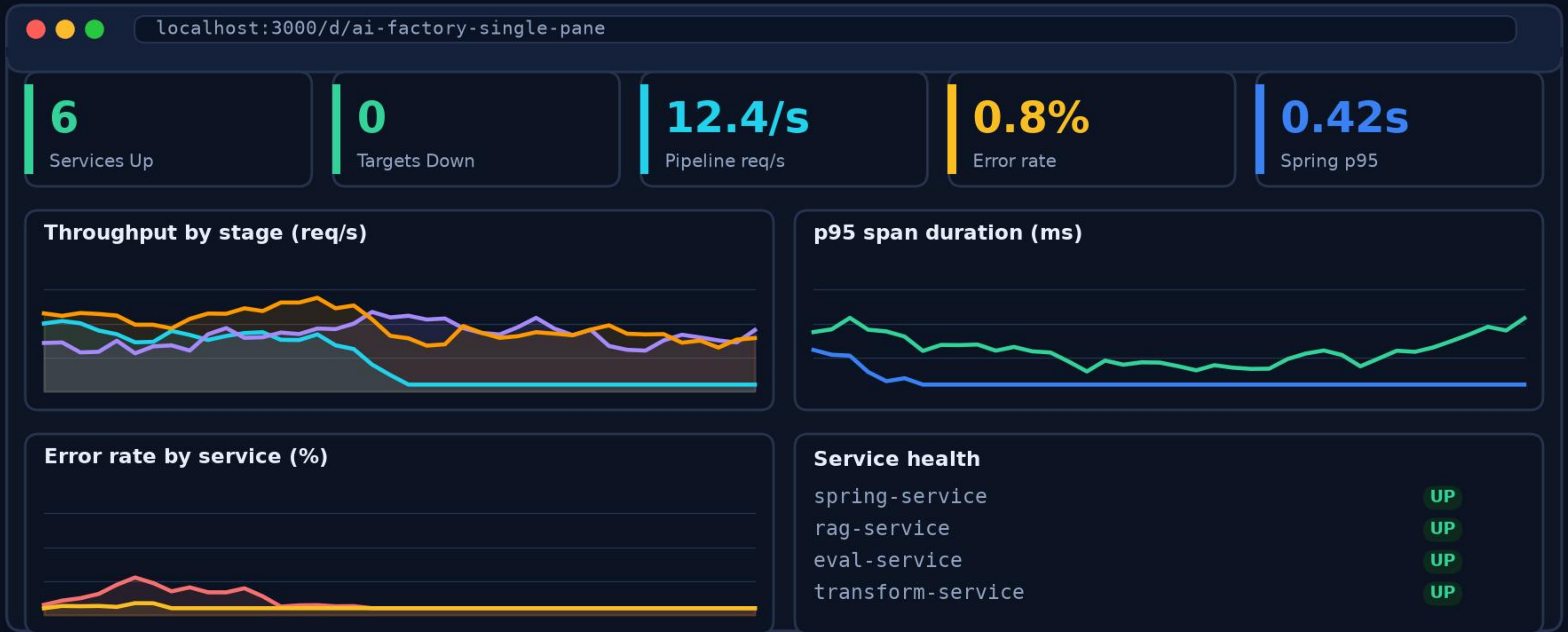
- RestTemplate, blocking I/O
- Solr search client
- Manual prompt strings
- No retries / resilience
- No telemetry, no eval
- CVE-bearing dependencies



MODERN · Spring Boot 3 + Spring AI 1.0

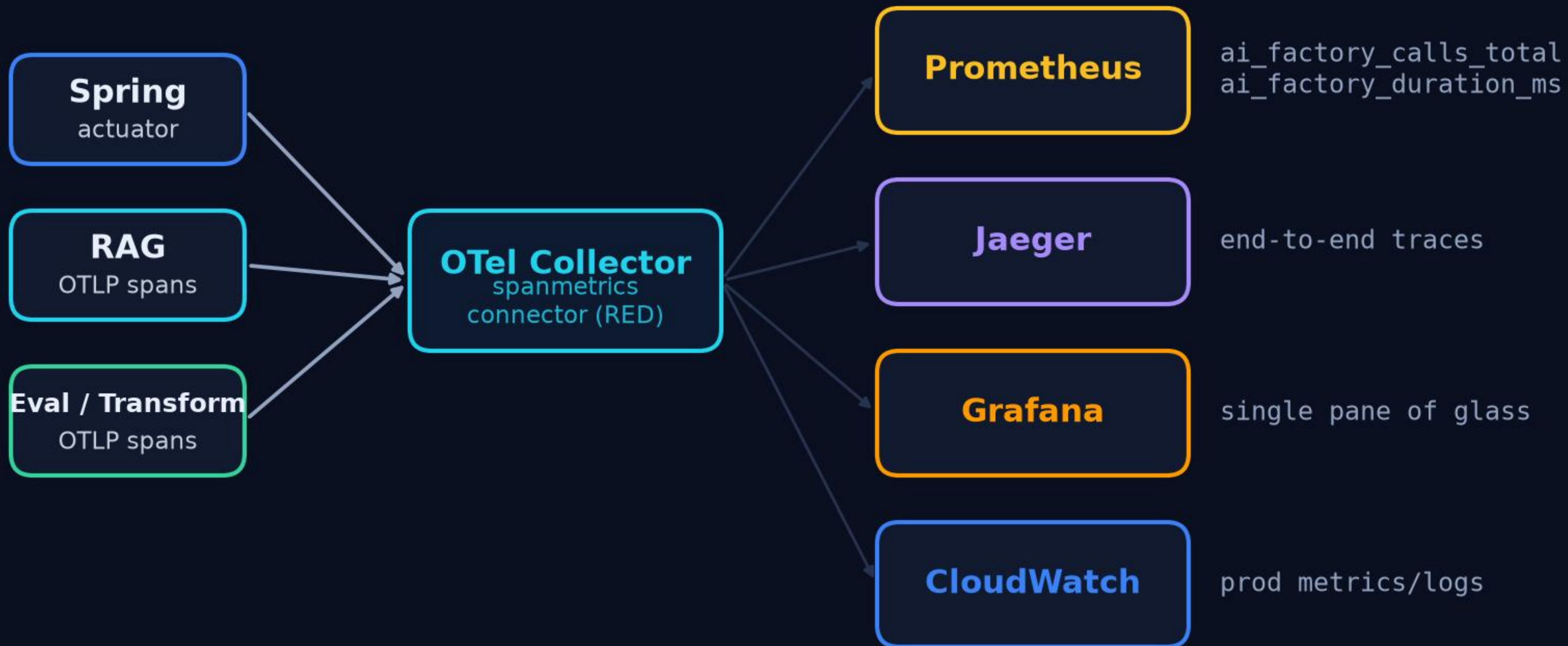
- WebClient, reactive + Resilience4j
- AnthropicChatModel (Spring AI)
- Grounded prompts from RAG
- OTel spans + metrics
- Eval gate before response
- CVE-remediated, 0 HIGH/CRITICAL

Single pane of glass — Grafana



Representative mockup — live dashboard needs `docker compose up` (spanmetrics RED metrics + Spring actuator).

Observability pipeline



Fix shipped: spanmetrics turns the Python services' traces into RED metrics so every stage shows up in Grafana.

AWS deployment architecture

VPC · per environment (dev / sit / preprod / prod)



- One Terraform stack, four env var-files; isolated S3 state per env
- Right-sized per environment; canary on preprod/prod

CI/CD with Harness — gated promotion

CI · build + scan



CD · promote



Promotion gates

- Smoke tests (dev) → ZAP DAST clean + approval (sit)
- Canary verified + approval (preprod) → 2-person approval + canary (prod)
- Auto-rollback on canary failure (25% → 10-min verify → 100%)

Walkthrough — bring it up & ingest



bash — quick start

```
→ docker compose up -d
✓ 10 services healthy (rag, eval, transform, spring, ...)

→ python scripts/seed_data.py --source ./data/sample-docs/
Seeded 42/42 chunks ✓

→ python scripts/ingest_sources.py --list
Available sources: confluence, sharepoint

→ python scripts/ingest_sources.py --source confluence
Ingesting from 'confluence' → http://localhost:8001
Ingested 2 documents → 3 chunks [sample (offline fallback)] ✓

→ python scripts/ingest_sources.py --source sharepoint
Ingested 2 documents → 4 chunks [sample (offline fallback)] ✓
```

What just happened

- Full local stack on Docker
- Sample corpus seeded into RAG
- Confluence + SharePoint connected
- Offline fallback → no creds needed
- Index now holds wiki + sample docs

Set CONFLUENCE_* / SHAREPOINT_*
for live ingestion

Query → grounded, gated answer

```
bash — query
→ curl -X POST localhost:8001/query \
→   -d '{"query":"What is our Q2 revenue forecast?"}'

{
  "answer": "The Q2 APAC revenue forecast
            is AUD 4.2M, based on the
            confirmed pipeline.",
  "sources": [{"title": "Q2 Forecast",
                  "url": "confluence/.../100021"}],
  "sufficiency_score": 0.91,
  "hops": 2,
}
```

Eval gate result

Answer relevancy	≥ 0.80	0.93
Faithfulness	≥ 0.85	0.90
Contextual recall	≥ 0.70	0.81
Hallucination	≤ 0.10	0.03
Toxicity	≤ 0.05	0.00

✓ **PASS — answer released to user**

Quality, testing & security

99

automated tests

unit + integration, all green

0

ruff lint issues

consistent, clean code

100%

mocked I/O

tests need no infrastructure

0

HIGH/CRITICAL CVEs

gated by remediation flow

Testing strategy

- External services stubbed in conftest
- DeepEval suite gated, not run by default
- Connector parsing covered by 24 tests
- Eval gate doubles as a CI quality gate

Security posture

- SAST (Semgrep, Gitleaks, OWASP) in CI
- Trivy image scanning before ECR push
- OWASP ZAP DAST against SIT
- PII redaction in the telemetry pipeline

Roadmap & next steps

- More connectors: Jira, Google Drive, S3, Slack
- Graph enrichment on ingest (entities → Neo4j)
- Incremental / scheduled sync with change detection
- Per-source access control + row-level filtering
- Alerting on the single-pane SLOs (error budget burn)

Delivered this iteration

- Single pane of glass + RED metrics
- Confluence & SharePoint connectors
- /ingest API + CLI, 24 new tests
- Test-collection fix (99 green)

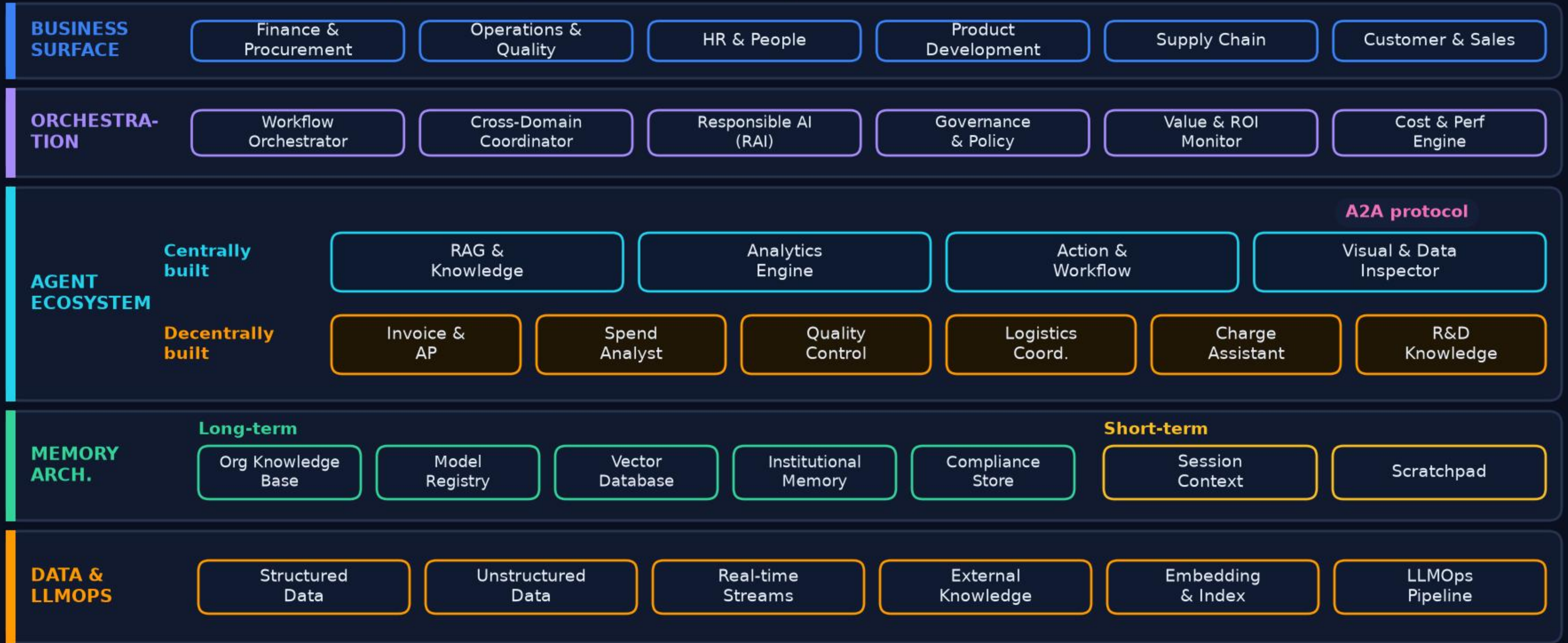
Next: scaling into a Cognitive Enterprise Platform →

Federated agents · shared memory · A2A orchestration

TARGET ARCHITECTURE

The Cognitive Enterprise Platform

Centralised + decentralised agents · long/short-term memory · A2A orchestration · built bottom-up



Build sequence (bottom-up): data foundation → memory → agents → orchestration → business value

Value streams across every function

Agents meet people where the work happens — one platform, many domains

Finance & Procurement

- Invoice & AP automation
- Spend & savings analyst
- Contract Q&A

Operations & Quality

- Quality control agent
- SOP / runbook copilot
- Incident triage

HR & People

- Onboarding assistant
- Policy & benefits Q&A
- Talent matching

Product Development

- R&D knowledge agent
- Design / code copilot
- Spec drafting

Supply Chain

- Demand forecasting
- Supplier risk watch
- Logistics coordinator

Customer & Sales

- Charge / billing assistant
- Proposal generator
- Next-best-action

The AI Centre of Excellence

Central control plane that coordinates, governs and optimises every agent



Workflow orchestration

Plan, route and supervise multi-step, multi-agent work end to end



Cross-domain coordination

Hand off context safely between function-specific agents



Responsible AI (RAI)

Guardrails, bias & safety checks, human-in-the-loop policies



Governance & policy

Access control, audit trails, model & data approvals



Value & ROI monitoring

Track outcomes, adoption and business value per agent



Cost & performance engine

Model routing, caching and token budget guardrails

Centrally built · decentrally built

A federated model: a paved-road platform that domain teams build on

Centrally built

Platform / CoE team

- Shared agents: RAG, analytics, action, inspector
- Reusable tools, memory & guardrails
- Golden datasets and eval harness
- Security, observability, FinOps baked in

Decentrally built

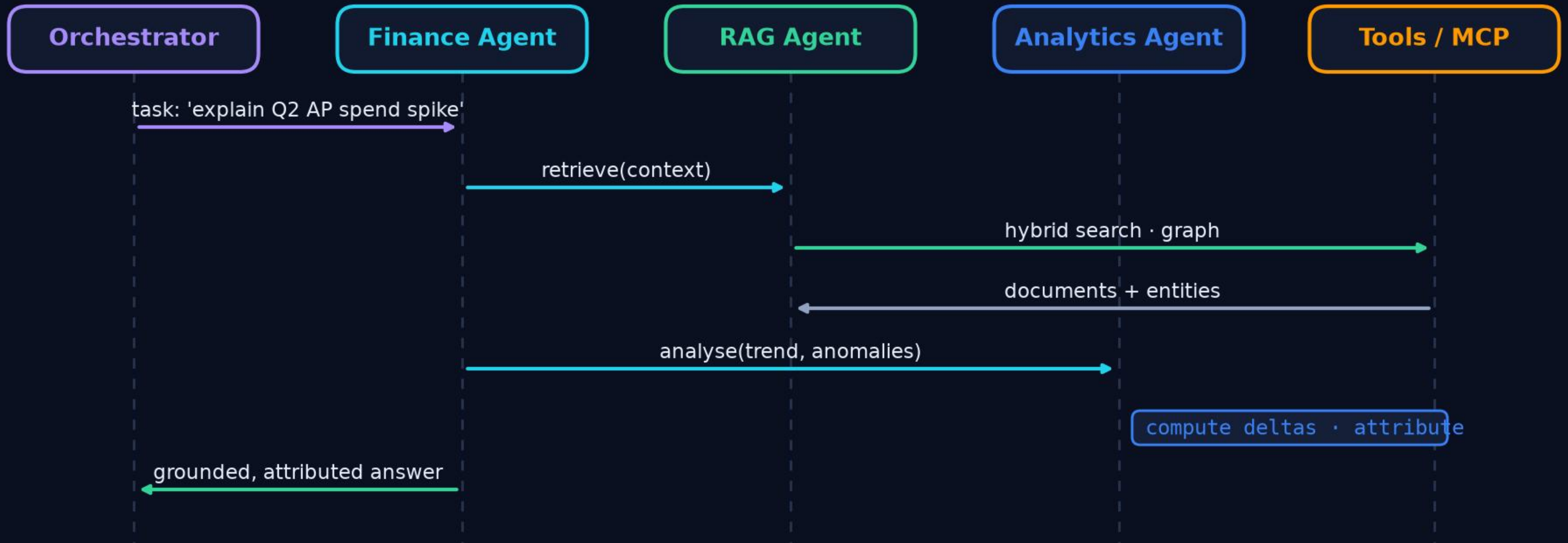
Business / domain teams

- Domain agents: invoice, quality, logistics...
- Own their prompts, data and use cases
- Move fast on the paved road
- Compose central agents via A2A

The paved road

A2A protocol Shared memory Guardrails Eval gates Observability Cost controls

Agent-to-Agent (A2A) collaboration



A2A lets specialised agents delegate sub-tasks with shared context — orchestrated, observable and policy-checked.

Memory architecture — what the platform remembers

Long-term memory

- **Org knowledge base** docs, wikis, policies → RAG index
- **Vector database** embeddings for semantic recall
- **Knowledge graph** entities & relationships (Neo4j)
- **Model & prompt registry** vetted, governed assets
- **Institutional memory** past decisions, outcomes, feedback
- **Compliance store** audit logs, lineage, approvals

Short-term memory

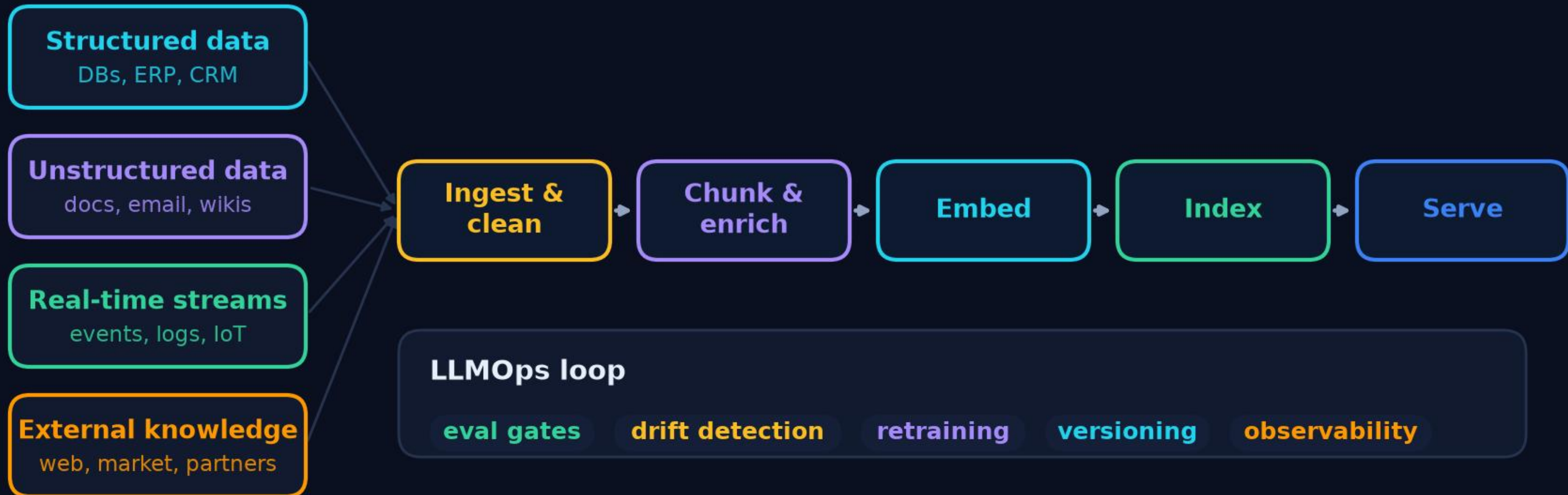
- **Session context** current conversation & goals
- **Scratchpad** intermediate reasoning & plans
- **Tool results cache** reuse within a single task
- **Working set** retrieved chunks in play

Short-term memory is cleared / summarised at task end — and promoted to long-term when it proves valuable.

FOUNDATION

Data foundation & LLMOps

From raw enterprise data to a production, observed pipeline



Bottom-up: a solid data + LLMOps base is what makes agents reliable, current and safe in production.

How this spike maps to the platform

Platform layer	Delivered in this spike	Next step
Business surface	Spring API + query endpoint	Per-function agent UIs
Orchestration	Kiro 5-stage agent spec	Multi-agent A2A orchestrator
Agent ecosystem	RAG / eval / transform agents	Decentralised domain agents
Memory	OpenSearch vectors + Neo4j graph	Institutional & session memory
Data & LLMOps	Connectors, seed, eval gates	Streaming + drift + retrain
Governance	Eval gates, PII redaction, SAST/DAST	RAI policies, value/ROI monitor

Security, guardrails & Responsible AI

Input guardrails

- Prompt-injection detection
- PII / secret screening
- Access & entitlement checks

Output guardrails

- Groundedness / citation check
- Toxicity, bias, safety filters
- Eval gate before release

Platform controls

- RBAC + audit trails
- Data lineage & approvals
- Human-in-the-loop for high risk

Defence in depth

Every request passes input → policy → retrieval → generation → output gates, with full telemetry and audit at each hop.

Cost & performance engine

Make agents fast and affordable at scale

-62%

cost per query
via caching + routing

38%

cache hit rate
semantic + exact

3

model tiers
route by difficulty

100%

budget-guarded
hard token caps

Levers

- Route easy calls to small models
- Cache embeddings & frequent answers
- Batch + stream to cut latency
- Trim context with normalisation

Governance

- Per-team token budgets & alerts
- Cost attributed by agent & domain
- ROI tracked against business value
- Anomaly detection on spend

Personas & journeys



Knowledge worker

Asks a question

Gets a grounded, cited answer in seconds



Domain builder

Builds an agent

Composes platform agents on the paved road



Platform engineer

Operates the platform

Ships shared agents, memory & guardrails



Risk & compliance

Governs usage

Audits, approves and monitors every agent

From pilot to platform



We are here: Walk — connectors, single pane of glass and eval gates are live.

Implementation roadmap

Now · 0-3 mo

- Harden RAG + connectors
- Single pane of glass SLOs
- 2-3 lighthouse use cases
- Eval gates in CI/CD

Next · 3-9 mo

- A2A multi-agent orchestrator
- Institutional + session memory
- Domain agent toolkit
- FinOps cost engine

Later · 9-18 mo

- Federated agent marketplace
- Streaming + drift + retrain
- Self-optimising routing
- Org-wide rollout

Business value & ROI

3-5x

faster knowledge retrieval

40-60%

less manual rework

weeks→days

legacy modernisation

24/7

consistent, audited answers

Value levers

● **Productivity**
agents handle the busywork

● **Quality**
grounded, gated, consistent

● **Speed**
modernise & ship faster

● **Risk**
governed, auditable, safe

● **Cost**
routing + caching + budgets

● **Scale**
one platform, many domains

From spike to Cognitive Enterprise Platform

Grounded · governed · observable · cost-aware AI — built bottom-up, scaled org-wide.

Agentic RAG · A2A orchestration · Shared memory · Eval gates · FinOps

Call to action

Pick 2-3 lighthouse use cases · stand up the paved road · prove ROI · then federate across the business.

Thank you · Questions & live demo